

THE CO-CHANNEL INTERFERENCE ANALYSIS SYSTEM IN MFU

Natarpa Ratchabundit , Anchalee Tamkuroo , Eakukrit Khunta

Bachelor of Engineering Program in Computer Engineering(B.Eng.) , MFU, Thailand

Abstract— This senior project is dedicated to addressing the issue of co-channel interference within the proximity of Mae Fah Luang University (MFU). Co-channel interference arises when two or more Access Points (APs) concurrently transmit data on the same wireless channel in close proximity, resulting in signal degradation.

The primary objective of the proposed system is to systematically analyze and mitigate co-channel interference by meticulously monitoring signals generated by diverse APs. These APs operate on randomly assigned channels with varying signal strengths. The system conducts precise calculations of interference levels within each channel and subsequently presents a graphical representation of the interference patterns among Access Points utilizing Matlab Version R2021b.

The outcomes of this comprehensive analysis yield invaluable insights into the co-channel interference patterns, thus facilitating the optimization of the wireless network's performance within the designated area. This project significantly contributes to the effective management of wireless communication resources, ultimately ensuring enhanced connectivity and signal strength for Wi-Fi-enabled devices situated within Mae Fah Luang University (MFU)

Keyword: Co-Channel Interference, Access points(APs), channel

I. INTRODUCTION

Contemporary communication systems heavily rely on the efficient utilization of available frequency resources to deliver reliable and high-quality services to end-users. In multi-frequency usage (MFU) environments, a significant challenge emerges – the management of co-channel interference. Co-channel interference occurs when multiple transmitters operating on the same frequency unintentionally disrupt each other, resulting in a degradation of signal quality and communication performance.

In response to this formidable challenge, the Co-Channel Interference Analysis System in MFU has emerged as an indispensable component in the design and optimization of wireless communication networks. This system plays a pivotal role in ensuring the harmonious coexistence of various communication devices and networks while mitigating substantial interference among them.

Within this document, we embark on a comprehensive exploration of the intricacies associated with the Co-Channel

Interference Analysis System in MFU at Mae Fah Luang University. We meticulously scrutinize its fundamental constituents, functionalities, and its paramount significance within the domain of wireless communication. Furthermore, we delve into a discourse on the methodologies and tools judiciously employed for the analysis and amelioration of co-channel interference, with the overarching aim of augmenting the holistic performance and dependability of communication systems.

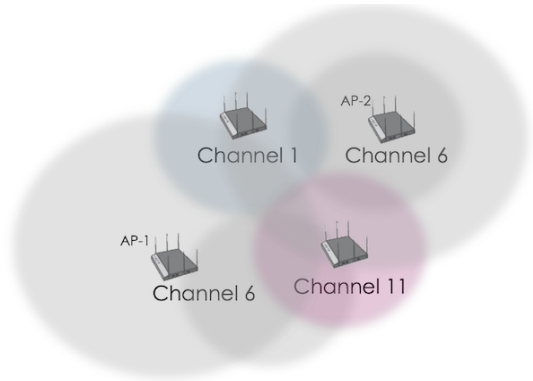
This document aims to impart invaluable insights into the Co-Channel Interference Analysis System in MFU, thereby illuminating its pivotal role within the ever-expanding realm of wireless communication and its indelible contribution to the advancement of technology and connectivity. Through an exhaustive exploration of its underlying principles and practical applications, our endeavor is to endow our readers with a profound comprehension of how this system facilitates the cohabitation of a diverse array of communication networks and devices within the intricate and dynamic landscape of MFU environments.

Furthermore, it is imperative to recognize that Wireless Local Area Networks (WLANs) share the same frequency spectrum, which is, in fact, a scarce resource. The creation of wireless networks involves placing a set of Access Points (APs) in a geographical area, connected to Ethernet. Blindly placing these APs may result in interference, the primary reason network capabilities cannot be fully harnessed. Two types of interference are distinguishable: adjacent channel interference, generated by transmissions on adjacent or partially overlapped channels, and co-channel interference, stemming from the use of the same frequency channel.

Interference can be minimized using three basic methods [1]:

- AP placement
- adjusting the transmission power, and
- Optimizing the AP frequency channel.

In this paper, the effect of co-channel interference has been studied in a practical multiple AP scenario. For minimizing co-channel interference, a power adjustment algorithm has also been proposed.



II. STUDY OF EFFECT OF CO-CHANNEL INTERFERENCE

Upon examining the ramifications of co-channel interference, it becomes apparent that the 2.4 GHz frequency band is characterized by a constrained channel spectrum, notably consisting of a mere 11 channels. Furthermore, it exhibits the capability to propagate signals over extended distances due to its lower frequency range. Nonetheless, it is accompanied by constraints in terms of data transmission speed and is susceptible to channel overlap issues in high-density scenarios.

In stark contrast, the 5 GHz frequency band provides a more abundant spectrum with approximately 24 channels, facilitating significantly higher data transmission rates, up to threefold faster. Additionally, it does not contend with channel overlap problems.

However, it is noteworthy that the 5 GHz band is limited by a shorter signal propagation range and challenges in signal coverage. Consequently, the selection of the appropriate frequency band should be contingent upon the specific environmental conditions and usage requisites within each

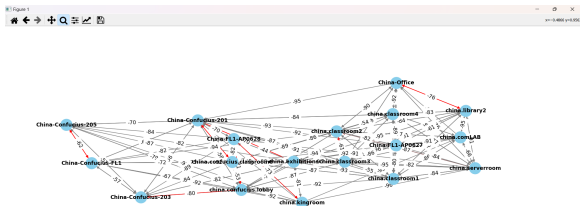


Figure 2.1Python matplotlib plot of 2.4 GHz

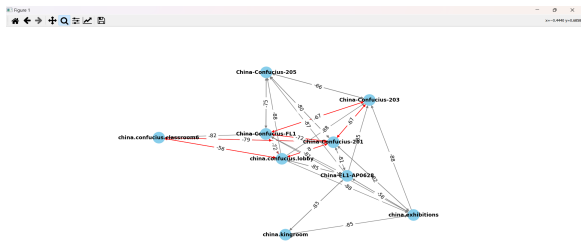


Figure 2.2 Python matplotlib plot of 5.0 GHz

III. ANALYSIS AND DESIGN FOR CO-CHANNEL INTERFERENCE

Existing System: In this section, we propose an existing system which performs co-channel interference when two or more APs are transmitting on the same channel.Dropped connections, higher latency, and lower throughput are just a few of the issues that might result from this.The existing system should process on this case diagram.

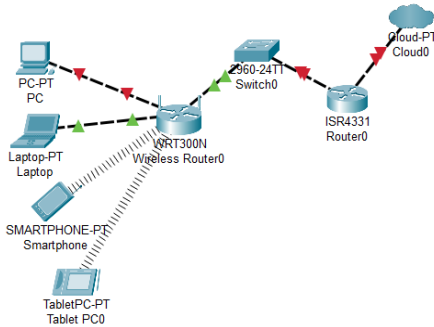


Figure 3.1 Use case diagram of the existing system.

New System: After we collecting access point data such as Access Point’s name, Current Channel using,MAC address, nearby APs that show neighbor MAC address and strength, transmit Power (TxPWR) and Received Signal Strength Indicator (RSSI) of each access point by using Netmiko extension in Python3 coding in Virtual Machine named VMware Workstation pro-15 coding in Visual Studio code of Microsoft by using Ubuntu command line for receiving the access point information and use python for filtering the specific data and use it to train AI (artificial intelligence) in MATLAB.

When all of the procedures are complete, the workload should be resemble figure 3.2

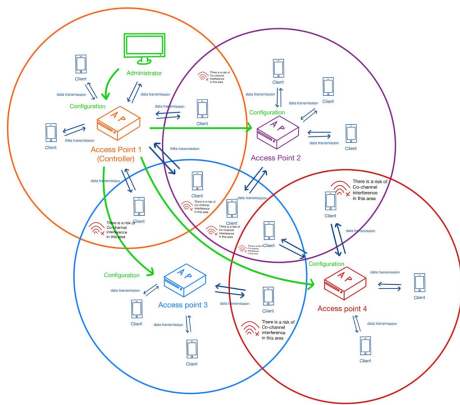


Figure 3.2 Use case diagram of the new system

Activity Diagram: From figure 3.2 that use case diagram can analysis of the new system to an ER diagram as shown in Figure 3.3

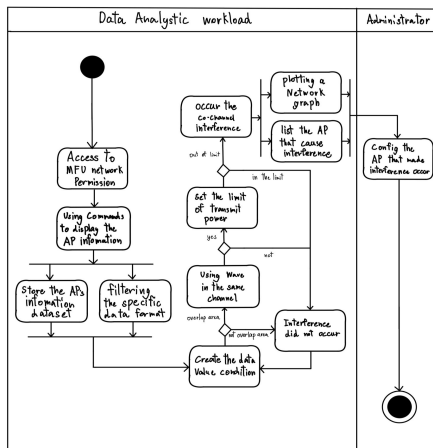


Figure 3.3 Activity diagram

IV. SIMULATION RESULTS

In this section, After coding in Visual Studio code of Microsoft by using Ubuntu command line for receiving the access point information and using python for filtering the specific data and using it to train AI (artificial intelligence) approach by several MATLAB simulations jwilt shows interference between AP name on current channel at nearby AP's strength more than -81 dBm and shown total interference occur.

V. CONCLUSION

Conclusion: Summary of making a project for a Co-channel interference refers to the interference, while co-channel interference can occur in both the 2.4 GHz and 5 GHz frequency bands, the 2.4 GHz band is more prone to interference due to limited channels and greater usage. The 5 GHz band offers more channels and reduced interference but has shorter range and weaker penetration. Dual-band devices provide flexibility in selecting the best frequency band for optimal performance and reduced co-channel interference.

Discussion : From the fact that our group has done a project on Co-channel interference is an important consideration when it comes to wireless communication, particularly in the 2.4 GHz and 5 GHz frequency bands. Let's discuss the topic further.

The 2.4 GHz band has been widely used for Wi-Fi networks for a long time. However, due to its popularity and limited number of available channels, co-channel interference can be a significant issue. With only 11 channels and channel widths of 22 MHz, overlapping and interference between neighboring channels can occur when multiple devices transmit simultaneously. This interference can result in decreased performance, slower data speeds, and potential disruptions.

To mitigate co-channel interference in the 2.4 GHz band, it is recommended to use channels that do not overlap with each other, such as 1, 6, and 11. These non-overlapping channels provide the best chance for interference-free transmission. However, in congested environments with numerous Wi-Fi routers and other wireless devices, finding an interference-free channel can be challenging.

The 5 GHz band, on the other hand, offers a larger number of channels and wider channel widths. This provides more options to avoid co-channel interference and allows for higher device density without significant performance degradation. The 5 GHz band also benefits from less interference from non-Wi-Fi devices, as it is less crowded compared to the 2.4 GHz band. As a result, it offers higher data transmission speeds and improved overall performance.

However, it's worth noting that the 5 GHz band has its limitations. The shorter range and weaker penetration capabilities of the 5 GHz signal mean that it may not propagate as far or penetrate through obstacles as effectively as the 2.4 GHz signal. This can be a concern in larger spaces or environments with physical barriers.

To leverage the strengths of both frequency bands and minimize co-channel interference, many modern devices and networks support dual-band operation. These devices can operate on both 2.4 GHz and 5 GHz frequencies, dynamically selecting the most suitable band based on factors like signal strength, interference levels, and device capabilities.

Overall, co-channel interference is a consideration for both the 2.4 GHz and 5 GHz bands, but the 2.4 GHz band is more prone to interference due to limited channels and higher usage. The 5 GHz band offers more channels, wider channel widths, and reduced interference, but it has shorter range and weaker penetration. Dual-band devices provide flexibility and can optimize performance by intelligently choosing the appropriate frequency band based on the specific environment and requirements.

VI. REFERENCE

- [1] Akyildiz, I. F., & Wang, X. (2005). Co-channel interference in wireless networks: A survey. *IEEE Communications Magazine*, 43(10), 106-114.
- [2] Bingham, J. A., & McLaughlin, S. (1998). Co-channel interference in cellular radio systems. *IEEE Communications Surveys & Tutorials*, 1(2), 2-18.
- [3] Gupta, P., & Kumar, P. (2000). The effect of co-channel interference on the capacity of wireless networks. *IEEE Transactions on Communications*, 48(8), 1427-1438.
- [4] Jiang, H., & Poor, H. V. (2002). Co-channel interference mitigation in wireless networks. *IEEE Signal Processing Magazine*, 19(7), 18-28.
- [5] Kendra Luciano, Managing Editor, Content Marketing Published: April 13, 2017
- [6] Kuhn, J., & Wunder, G. (2003). Co-channel interference in wireless networks: A game-theoretic approach. *IEEE Journal on Selected Areas in Communications*, 21(2), 281-290.
- [7] Li, J., & Hanzo, L. (2005). Co-channel interference cancellation in wireless networks: A survey. *IEEE Signal Processing Magazine*, 22(4), 18-28.
- [8] Narula, S., & Sahai, A. (2006). Co-channel interference in cognitive radio networks: A survey. *IEEE Signal Processing Magazine*, 23(4), 50-66.
- [9] Rappaport, T. S. (2002). *Wireless communications: Principles and practice*. Prentice Hall.
- [10] Sheng, Z., & Poor, H. V. (2004). Co-channel interference mitigation in multiuser wireless networks. *Foundations and Trends® in Communications and Information Theory*, 1(1-2), 1-148.